

# Kuldeep Tanwar

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Portfolio | LinkedIn | GitHub | LeetCode | GFG

## PROFESSIONAL SUMMARY

Full-stack Engineer with 2.5+ years of experience building **scalable, production-grade systems** using JavaScript, TypeScript, React, Next.js, Node.js, NestJS, FastAPI, Python, and AWS; **mentoring 2 interns**. Skilled in **backend architecture and performance optimization**, leveraging Redis caching, database indexing, query tuning, asynchronous processing, and distributed system design to improve scalability and reliability. Experienced in **AI-powered product development** using Claude API, OpenAI, GPTZero, and LLM pipelines to automate workflows and enhance user experiences at scale. Proven ability to **own features end-to-end**, optimize system performance, and deliver business impact.

## EXPERIENCE

**Software Development Engineer 1** (*React, Next.js, Node.js, NestJS, TypeScript, MongoDB, AWS*) Nov 2024 – Present  
Digiaccel Learning | Altera Institute Gurugram, IN

- Built a **Grafana/Prometheus** observability dashboard with **AWS Lambda**-driven staging shutdown, reducing infrastructure costs by **12.5%**.
- Reduced **API response times by 75%** via **Redis caching**, concurrent async processing, query optimization, and request batching.
- Established scaling baselines through **k6 load tests** on **AWS ECS Fargate**, informing peak-traffic auto-scaling and capacity policies.
- Integrated **Sentry** for error tracking and **PostHog** for product analytics, cutting **MTTR to ~1 hour** and surfacing engagement insights.
- Implemented secure direct-to-**AWS S3** uploads via **pre-signed URLs**, removing server bottlenecks and improving upload latency by **~30%**.
- Applied **BullMQ** rate-limiting and retry queues on **LeadSquared CRM** lead-sync, eliminating HTTP 429 throttling with zero lead loss.

**Junior Software Engineer** (*JavaScript, React, Node.js, Express, MongoDB*) Dec 2023 – Nov 2024  
Vilihi Virtual Services Noida, IN

- Published a shared **TypeScript Enums NPM package** unifying frontend and backend contracts, cutting type-related defects by **~20%**.
- Configured **Storybook** for isolated React component development, accelerating UI iteration, design QA, and review cycles.
- Designed **Node.js + Express** REST APIs with **MongoDB** schemas, **JWT** auth, and **Zod** validation powering core backend workflows.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript (ES6+), Python, C++ , SQL

**Frontend:** React, Next.js, Redux, Tailwind CSS, MUI, HTML5, Progressive Web Apps (PWA), Storybook

**Backend:** Node.js, NestJS, Express.js, FastAPI, REST APIs, Microservices, WebSockets

**Databases & Caching:** MongoDB, MySQL, PostgreSQL, Redis, BullMQ, Celery

**AI & Integrations:** Claude API, OpenAI APIs, GPTZero, LLM Pipelines, Meta Graph API, Webhooks, Prompt Engineering

**Cloud & DevOps:** AWS (S3, ECS, Fargate, Lambda, RDS, SQS, SES, CloudWatch), Docker, Nginx, CI/CD, Vercel, Sentry, PostHog, k6

**Tools & Practices:** Git/GitHub, JIRA, Postman, Agile/Scrum, System Design, Code Review

## KEY PROJECTS

**DigiCampus** (Live) | *Learning Management System — React, Next.js, Node.js, NestJS, JavaScript, TypeScript, MongoDB, AWS, Redis*

- Built and scaled a production LMS serving **5,000+ learners** and facilitating **150k–300k learning hours monthly** across multiple cohorts.
- Cut **API latency by 75%** via **Redis caching**, concurrent async processing, and strategic **MongoDB indexing** on the hottest read paths.
- Integrated **GPTZero** AI APIs to detect AI-generated content in student assessment submissions, mitigating cheating at scale.
- Converted the platform into a **Progressive Web App** with offline access, install support, and push-notification engagement.
- Optimized React with **memoization**, **code-splitting**, **lazy loading**, **debouncing**, and AbortController-based request cancellation patterns.

**Narriva** (Website — Play Store) | *AI Social Media Automation — Python, FastAPI, PostgreSQL, Redis, Celery, React Native, AWS, Claude API*

- Architected an **AI-driven social media automation** platform for content generation, publishing, customer interaction, and lead capture.
- Built **event-driven async pipelines** on **Celery + Redis** for trend detection, scheduling, lead capture, and WhatsApp alerts.
- Implemented **Meta Graph API** for Instagram/Facebook **OAuth**, media publishing, real-time **webhook** processing, and analytics.
- Engineered **FastAPI** services with **PostgreSQL**, **AWS**, **JWT** auth, and rate-limited, multi-tenant request handling at scale.
- Designed scalable **SaaS architecture** for SMB automation with secure APIs, background jobs, and third-party platform integrations.

## EDUCATION

**National Institute of Technology Kurukshetra** 2020 – 2024  
*Bachelor of Technology* CGPA: 7.88/10

**Yaduvanshi Shiksha Niketan, Mahendragarh** 2018 – 2020  
*Senior Secondary (Class XII)* 93.2%

**Yaduvanshi Shiksha Niketan, Mahendragarh** 2016 – 2018  
*Secondary (Class X)* 88.5%